NOT MEASUREMENT SENSITIVE

MIL-STD-361-10(TM)
15 JULY 1992

SUPERSEDING
(See 6.5.)

MILITARY STANDARD

MANUALS, TECHNICAL:

LUBRICATION INSTRUCTIONS

AMSC A6759

AREA TMSS

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FOREWORD

1. This military standard is approved for use by the Department of the Army and is available for use by all Departments and Agencies of the Department of Defense (DOD).

2. Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander USAMC Materiel Readiness Support Activity, ATTN: AMXMD-EP, Lexington, Kentucky 40511-5101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of the document or by letter.

3. This document is one part of an 11-part bookform standard to be used in the preparation and acquisition of technical manuals that are essential to support (operate and maintain) the various types of equipment and weapon systems within the Department of the Army.

4. This document covers the lubrication instruction requirements for the preparation of technical manuals and lubrication orders.

5. Other parts of this multi-part standard are:

MIL-STD-361-1	-	Writing Style and Format Requirements.
MIL-STD-361-2	-	Comprehensibility Requirements.
MIL-STD-361-3	-	Requirements for Graphics.
MIL-STD-361-4	-	Quality Assurance Requirements.
MIL-STD-361-5	-	Requirements for Packaging.
MIL-STD-361-6	-	Introductory Information with Theory of Operations Requirements.
MIL-STD-361-7	-	Operator and Preventive Maintenance Instructions.
MIL-STD-361-8	-	Maintenance Instructions.
MIL-STD-361-9	-	Troubleshooting Procedures.
MIL-STD-361-11	-	Repair Parts and Special Tools List (RPSTL).

6. By incorporating standardized and service-unique technical manual requirements, this document and its dash-numbered companion parts, along with the general acquisition specification MIL-M-63008, Acquisition/Preparation Requirements, support the acquisition of technical manuals.

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1. SCOPE

1.1 Scope. This document establishes the content and procedural requirements for preparation of lubrication instructions for technical manuals (TMs), revisions, supplements, and changes.

1.2 Purpose. This document supplements the various content requirements in the other parts to this 11-part standard for specific types of technical manuals and related publications and does not alone deliver any technical data. (NOTE: The figures used in this document are examples only. The text of this document takes precedence over the figures.)

1.3 Applicability. The applicability of the requirements contained in this document is governed by the equipment, systems, or weapon systems for which the technical manuals or lubrication instructions are being developed. This document is applicable for use by the Department of the Army.

1.4 Selective application and tailoring. This document contains some requirements that may not be applicable to the preparation of all technical manuals and lubrication instructions. Selective application and tailoring of requirements contained in this document are the responsibility of the contracting activity and shall be accomplished through the use of MIL-M-63008. The applicability of some requirements is also designated by one of the following statements:

- unless requirement is specifically excluded by the contracting activity; or
- as/when specified by the contracting activity.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation. (See 6.2.)

SPECIFICATIONS

MILITARY

MIL-M-63008	-	Manuals, Technical: Acquisition/ Preparation Requirements.

STANDARDS

MILITARY

MIL-STD-361-1	-	Manuals, Technical: Writing Style and Format Requirements.
MIL-STD-361-2	-	Manuals, Technical: Comprehensibility Requirements.
MIL-STD-361-3	-	Manuals, Technical: Requirements for Graphics.
MIL-STD-361-4	-	Manuals, Technical: Quality Assurance Requirements.
MIL-STD-361-5	-	Manuals, Technical: Requirements for Packaging.
MIL-STD-1388-2	-	DOD Requirement for a Logistic Support Analysis Record.

HANDBOOKS

MILITARY

MIL-HDBK-113	-	Lubricants, Power Transmission Fluids, Corrosion Preventatives for Use in Ground Equipment Systems, Guide for Selection of.
MIL-HDBK-275	-	Guide for Selection of Lubricants, Fluids, and Components for Use in Flight Vehicles and Components.

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(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from Standardization Documents Order Desk, Bldg 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5904.)

2.1.2 Other Government document(s) and publication(s). The following other Government document/publication forms a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

TB 750-93-1	-	Functional Grouping Codes: Combat, Tactical, and Support Vehicles and Special Purpose Equipment.

(Copies of other Government documents and publications required by the contractor in connection with specific acquisition functions should be obtained from the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this document ("This document" includes all volumes of MIL-STD-361 series.) and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. DEFINITIONS

3.1 Acronyms used in this standard. The acronyms used in this standard are defined as follows:

a. AOAP - Army Oil Analysis Program.

b. DOD - Department of Defense.

c. DODISS - Department of Defense Index of Specifications and Standards.

d. LSA - Logistic Support Analysis.

e. LSAR - Logistic Support Analysis Record.

f. MAC - Maintenance Allocation Chart.

g. NSN - National Stock Number.

h. RCM - Reliability Centered Maintenance.

i. TM - Technical Manual.

3.2 Army oil analysis program (AOAP). Effort to detect impending equipment component failure and determine lubricant condition through periodic analytical evaluation of oil samples.

3.3 Contracting activity. The Department of Defense (DOD) component, activity, or organization of a using military service or that organization delegated by a using service, that is responsible for the selection and determination of requirements for TMs.

3.4 Hardtime scheduled maintenance. Hardtime maintenance is scheduled maintenance conducted at predetermined fixed intervals because of age, calendar, or usage such as operating time, flying hours, miles driven, or rounds fired.

3.5 Lubricant. Any solid, fluid, or semifluid material that performs a lubricating or related specialty function. Such materials include lubricating oils, greases, hydraulic fluids, damping fluids, dielectric coolants, antiseize compounds, corrosion preventatives, and bonded or unbonded solid films.

3.6 Maintenance repair action. Lubrication task required only when a specified maintenance repair action is being performed.

3.7 On-condition scheduled maintenance. On-condition scheduled maintenance is maintenance or item replacement action performed based upon condition of the item as determined by an evaluation of each item on a scheduled basis.

3.8 Operator maintenance. Operator maintenance consists of inspecting, servicing, lubricating, adjusting, replacing, and

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repairing those items authorized by logistic support analysis (LSA) and/or maintenance allocation chart (MAC).

3.9 System. A regularly interacting or interdependent group of items forming a unified whole to perform a specific operational function.

3.10 Technical manuals (TMs). Manuals that contain instructions for the installation, operation, maintenance, and support of weapon systems, weapon system components, and support equipment. TM information may be presented, according to prior agreement between the contractor and the Government, in any form or characteristic, including hard printed copy, audio and visual displays, electronic imbedded media, disks, other electronic devices, or other media. They normally include operational and maintenance instructions, parts lists, and related technical information or procedures exclusive of administrative procedures.

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4. GENERAL REQUIREMENTS

4.1 General style requirements.

4.1.1 Conformance to standard. The general manner for the preparation of the cover/title page, front matter, text, security classification, classification markings, numbering, changes, warnings, cautions, health hazards, notes, final reproducible copy of final TM, and other document elements shall be in accordance with MIL-STD-361-1.

4.1.2 Technical content preparation. The following baseline requirements shall be used as specified by the contracting activity:

a. Logistic support analysis record (LSAR). The technical data and instructions developed by the requirements of MIL-STD-1388-2, DOD Requirements for a Logistic Support Analysis Record (LSAR), shall be used as the baseline to prepare TMs.

b. Maintenance allocation chart (MAC). The MAC shall be used as the baseline to prepare TMs. The MAC shall be prepared from the equipment top-down breakdown to consolidate and identify those groups on the list which involve identified maintenance functions. Logistic Support Analysis (LSA) Output Report 04, Maintenance Allocation Summary will be used when available as source data for the final approved MAC.

When specified by the contracting activity and if a waiver of standard requirements is granted, TB 750-93-1 may be used as the baseline for preparing TMs.

4.1.3 Security classification markings. All classified material shall be safeguarded and identified in accordance with MIL-STD-361-1.

4.2 Lubrication instructions. Lubrication instructions shall provide all essential information required by user personnel for performing all operator/unit, aviation unit, direct support, aviation intermediate, and general support maintenance lubrication on equipment, systems, and weapon systems. Lubrication procedures shall include detailed cleaning, inspection, and lubrication instructions to enable the user to receive, lubricate, and return to an acceptable performance standard all components of the equipment in a minimum of time with the skills, tools, test equipment, and spare parts authorized by the LSA/MAC. The method used for identification of lubrication points shall be explained. Reference (in accordance with the requirements of MIL-STD-361-1) to specific TMs or sections containing schematics and diagrams shall be acceptable.

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4.2.1 Development of text and graphics. Development of text and graphics shall be in accordance with the requirements of MIL-STD-361-1 and MIL-STD-361-3.

4.2.2 Lubrication under adverse conditions. Information shall be included on any special lubrication required under extreme temperature, altitude, and humidity conditions within the limits established by the design specification for the equipment.

4.2.3 Nomenclature. Nomenclature shall be in accordance with the requirements of MIL-STD-361-1.

4.2.4 Preparation of illustrations. Preparation of charts, tables, and lubrication diagrams shall be in accordance with MIL-STD-361-3.

4.2.4.1 Illustration keys. Legends, tables, or text applicable to partial page illustrations shall, whenever possible, appear on the same page. For full-page illustrations, the information shall appear on a facing page. If the legend, table, or text exceeds the allowable space, it shall be continued on a second page. For additional details, see MIL-STD-361-1 and MIL-STD-361-3.

4.2.4.2 Diagrams. The lubrication text for the TM shall include diagrams and associated data necessary for personnel to identify and accomplish general and specific lubrication requirements. The requirements of MIL-STD-361-2 and MIL-STD-361-3 shall be applicable. Any foldout diagrams required shall be prepared in accordance with MIL-STD-361-1.

4.2.4.2.1 Explanatory text for diagrams. An explanation of the diagrams used in the TM may be given and should include their content and proper use.

4.2.4.2.2 Symbols. The use of symbols shall be in accordance with MIL-STD-361-1.

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5. DETAILED REQUIREMENTS

5.1 General. This standard, along with the other companion documents of this standard, shall establish the style, format, and technical content requirements for the preparation of lubrication instructions for TMs.

5.1.2 Lubrication instructions. Lubrication instructions shall include all lubrication procedures and information on authorized lubricants, lubrication intervals, man-hour requirements, and the AOAP.

5.1.3 Reliability centered maintenance (RCM). Unless requirement is specifically excluded by the contracting activity, lubrication instructions shall be based upon the principles of RCM logic.

5.2 Security classification. Lubrication instructions that are classified as confidential or higher shall be published as a part of the associated TM of the same or higher classification. For additional instructions on security classification, see MIL-STD-361-1.

5.3 Format.

5.3.1 Grouped sequence (hardtime, on-condition). Unless requirement is specifically excluded by the contracting activity, the lubrication instructions shall be presented in grouped sequence by interval.

5.4 Content.

5.4.1 General statements. General statement(s) applicable to overall understanding of requirements shall be provided. The statement(s) shall include such information as adherence to lubrication intervals, explanation of interval symbols, maintenance levels, exceptional operational requirements, abbreviations, fittings, and parts cleaning. A statement concerning corrosion control shall be used as applicable. The statement shall provide instructions or reference corrosion control requirements provided in the applicable maintenance instructions.

5.4.1.1 Oil filter statement. As applicable, a statement similar to the following shall be included:

"Oil filters shall be serviced/cleaned/changed, as applicable, when:

a. They are known to be contaminated or clogged,

b. Service is recommended by AOAP laboratory analysis

or

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c. At prescribed hardtime intervals."

5.4.1.2 AOAP sampling interval statement. The following statement shall be included:

"Engine oil/transmission oil/hydraulic fluids must be sampled at (insert applicable hour/mileage timeframe) as prescribed by (insert TB 43-0106 or DA PAM 738-750)."

5.4.1.3 AOAP not available/non-enrolled statement. When a component/equipment is not enrolled in the AOAP or oil analysis support is not available, the following statement shall be used:

"This (enter name of component/equipment) is not enrolled in the Army Oil Analysis Program. HARDTIME INTERVALS APPLY."

5.4.1.4 Warranty hardtime statement. The following statement shall be used, as applicable:

"For equipment under manufacturer's warranty, hardtime oil service intervals shall be followed. Intervals shall be shortened if lubricants are known to be contaminated or if operation is under adverse conditions (such as longer-than-usual operating hours, extended idling periods, extreme dust)."

5.5 Illustrations. Illustrations shall be used to show the location of lubrication points and, as applicable, shall indicate the number of lubrication points. (See figure 1.) A minimum number of views shall be used.

5.5.1 Multiple illustrations. When it is necessary to provide a multiple number of illustrations to show separate component parts, each illustration shall have an individual title.

5.5.2 Grouped lubrication points. When grouped lubrication points require the same lubricant at the same interval, the type and number of points shall be identified and described by one of the following methods:

a. Multi-headed arrows. Multi-headed, solid-shafted arrows shall point to each of the lubrication points.

b. Lubrication point notes. Lubrication point notes shall provide instructions for applying lubricants, taking into account the following factors: type, grade, availability, and properties of the prescribed lubricant; expected temperature; lubrication guns and tools available to authorized maintenance level; types of lubrication fittings; and possible ill effects of excessive or insufficient lubrication. Caution shall be stressed where over- or under-lubrication of a part will damage that part or closely associated parts. Such cautionary notes shall be included either as a portion of the point note or as a special note.

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5.5.3 Disassembly and hand-packing. If applicable, disassembly and hand-packing instructions shall be provided for medium- and high-speed antifriction bearings that are sensitive to the amount of lubrication applied and do not have bleed holes or relief valves.

5.5.4 Cleaning, disassembling, and assembling. Cleaning, disassembling, and assembling instructions required before or after lubrication shall be provided. If instructions are extensive and contained in the maintenance instructions or a separate TM, the maintenance instructions or TM shall be referenced. Refer to MIL-STD-361-1 for referencing criteria.

5.5.5 Washing and natural drying. If applicable, instructions shall be given for washing and natural drying of finely machined and dirt-sensitive parts before relubricating. Use of compressed air jets or temperatures above 212°F shall not be prescribed.

5.5.6 Preservative material. Instructions shall not specify a coating of preservative material, either before or after packing parts that are lubricated with grease; nor shall they specify an application of oil, solvent, or additional grease to a "sealed-for-life" or prepackaged antifriction bearing.

5.6 Lubricants and military symbols. Lubricants shall be identified by standard military symbols in accordance with MIL-HDBK-113 and MIL-HDBK-275. These symbols shall be printed in uppercase letters in separate vertical columns on the inner side of the lubrication point names. These columns shall be headed by the words "LUBRICANT" and "INTERVAL." Those lubrication points that are serviced or lubricated by checking level, replenishing lubricant, or draining and refilling shall be indicated by the lubricant symbol at the point on the illustration that is designated for replenishing or refilling. The amount of lubricant required shall be given either in the point note or in the "CAPACITY" column of the lubricant table, if applicable.

5.6.1 Lubrication interval symbols. Unless requirement is specifically excluded by the contracting activity, the following lubrication interval symbols shall be used, as applicable:

D--daily
W--weekly
M--monthly
Q--quarterly
S--semiannually
A--annually
MRA--maintenance repair action
B--biennially
H--hours (operated)
MI--miles (operated)
KM--kilometers (operated)
RDS--rounds fired
OC--on-condition

5.7 Measurements. Unless requirement is specifically excluded by the contracting activity, all measurements expressed in text, in tables, or on illustrations shall be expressed in both US standard units and metric units. The order of precedence shall be in accordance with equipment markings.

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5.7.1 Lubricant table. As applicable, a table(s) shall be prepared to provide information needed to select the proper lubricant for various temperature ranges and uses. The size and location of the table(s) shall be tailored to meet layout requirements and shall include, as applicable, information on temperature range, lubricant, military symbol, NATO code, specification, national stock number (NSN), capacity, interval between service, and man-hours required to complete all services and shall be stated to the nearest tenth for all prescribed lubrication. (See figure 2.)

5.7.2 Notes to tables. As necessary, when specific restrictions, preferred grades, and other conditions exist, notes shall be annotated on table(s) in accordance with MIL-STD-361-1. For example: "When MIL-L-2104 lubricant is authorized, use 15W-40 (OE/HDO-15/40) when available and applicable temperature range exists," or "15W-40 oil is not authorized in this particular (enter component name)." Where applicable, the statement "For Arctic operation, refer to FM 9-207." shall be included as a note.

5.8 Special notes.

5.8.1 Pertinent lubrication point information. As applicable, additional pertinent lubrication point information shall be incorporated into a section titled "NOTES." When applicable, the instructions shall contain a special note referencing, but not repeating, instructions in maintenance chapters.

5.8.2 Effect of extreme humidity. If applicable, pertinent instructions relevant to the effect of extreme humidity on lubrication requirements for the equipment shall be included as a note.

5.9 Quality assurance. Quality assurance and quality control for TMs, revisions, supplements, and changes shall be in accordance with MIL-STD-361-4.

5.10 Packaging and preparation for delivery. TMs, revisions, supplements, and changes shall be packaged in accordance with MIL-STD-361-5.

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6. NOTES

(This section contains information of a general or explanatory nature that may by helpful, but is not mandatory.)

6.1 Intended use. This document prescribes the lubrication instructions requirements applicable to various types of TMs, revisions, supplements, and changes prepared by or for the Department of the Army.

6.2 Issue of DODISS. When this document is used in acquisition, the applicable issue of the DODISS must be cited in the solicitation. (See 2.1.1.)

6.3 Tailoring guidance. The contracting activity should tailor any required options offered herein in accordance with MIL-M-63008.

6.4 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this standard.

b. Issue of DODISS to be cited in the solicitation, and, if required, the specific issue of individual documents referenced. (See 2.1.1.)

6.5 Supersession data. This document consolidates, standardizes, and supersedes the lubrication instructions requirements contained in the following documents:

MIL-M-63004C(TM)	-	Manuals, Technical: Preparation of Lubrication Orders.
MIL-M-63036D(TM)	-	Manuals, Technical: Operator's, Preparation of.
MIL-M-63038C(TM)	-	Manuals, Technical: Organizational or Aviation Unit, Direct Support or Aviation Intermediate, and General Support Maintenance.

6.6 Subject term (key word) listing. The following terms are to be used to identify this document during retrieval searches:

Army oil analysis program (AOAP)
Content
Disassembly and hand-packing
Grouped lubrication points
Hardtime
Illustrations
Lubricant table
Lubricants and military symbols
Lubrication instructions

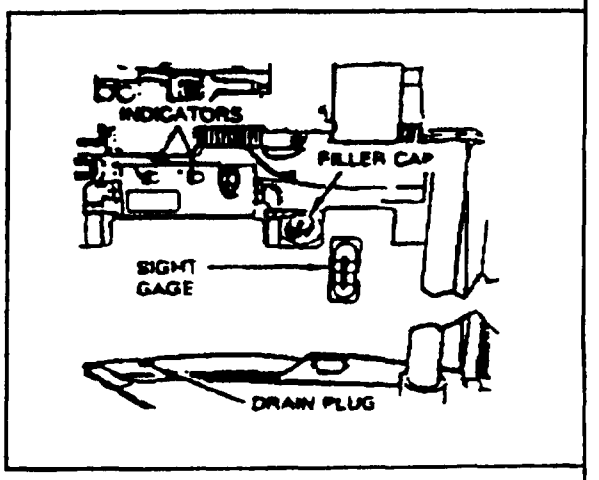
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Lubrication interval symbols
Notes to tables
On-condition
Packaging and preparation for delivery
Preservative material
Reliability centered maintenance (RCM)
Special notes
Washing and drying

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LUBRICANT - INTERVAL

HULL HYDRAULIC RESERVOIR OIL LEVEL (See Note 1.) (C)	FRH	D———
HULL HYDRAULIC RESERVOIR FILL (See Note 2.) (C)	FRH	•———



SUSPENSION SYSTEM

TOTAL MAN-HR	
Interval	Man-Hr.
D- Daily	0.1
•- Non-Scheduled/As Required	0.2

KEY

LUBRICANT/COMPONENTS	REFILL CAPACITY	EXPECTED TEMPERATURES*		
		Above +32°F	+40°F to -10°F	0°F to -65°F
FRH: Hydraulic Fluid, Rust Inhibited, Fire Resistant, Synthetic Hydrocarbon MIL-H-46170B Hull Hydraulic Reservoir	18 Gallons	ALL TEMPERATURES		

*For Arctic operation, refer to FM 9-207.

WARNING

FRH hydraulic fluid may contain tricresyl phosphate which, if taken internally, can produce paralysis. Hydraulic fluid may be absorbed through the skin. If hydraulic fluid gets on skin, thoroughly wash off with soap and water as soon as possible.

NOTES

1. HYDRAULIC RESERVOIR OIL LEVEL - Check fluid level daily and before operation. Perform hydraulic zero pressure check. (See TM 9-2350-265-10-2.) With both the hydraulic pressure gage located at the GPS and the parking brake hydraulic pressure gage located at the driver's station indicating zero psi, look at hydraulic reservoir sight gage. Fluid level should be above ADD 1 GAL mark. If not, add fluid. Visually check two indicators for evidence of filter bypass. If popped out, notify unit maintenance.

2. HYDRAULIC RESERVOIR FILL - To fill reservoir, traverse turret until basket opening faces reservoir. Clean area around filler cap with rag and solvent. Unscrew filler cap and add fluid, one gallon at a time, until fluid is above ADD 1 GAL mark. Install cap.

FIGURE 1. Example of identification of lubricant, symbol, and lubrication points, interval, and note (reference 5.5).

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Table X-X. Lubricant Table for Engine XXX.

TEMPERATURE RANGE	LUBRICANT MIL SYMBOL (NATO CODE) SPECIFICATION	NSN	CAPACITY	INTERVAL	MAN-HOUR
-18°C to +49°C (0°F to +120°F)	OE/HDO 15/40 (O-1236) MIL-L-2104	...	5 QT	200 MI	0.5
-25°C to +04°C (-15°F to +40°F)	OE/HDO-10 (O-237) MIL-L-2104	...	5 QT	200 MI	0.5
-10°C to +49°C (+15°F to +120°F)	OE/HDO-30 (O-238) MIL-L-2104	...	5 QT	200 MI	0.5
-05°C to +49°C (+25°F to +120°F)	OE/HDO-40 (NA) MIL-L-2104	...	5 QT	200 MI	0.5
-57°C to +04°C (-70°F to +40°F)	OEA (D-183) MIL-L-46167	...	5 QT	100 MI	0.5

FIGURE 2. Example of lubrication table (reference 5.7.1).

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